Logistic regression

David Puelz
The University of Austin

The OkCupid data set

- The `okcupid` data set contains information on 59,946 profiles from users of the OkCupid online dating service.
- We have user age, height, sex, income, orientation, education, body type, ethnicity, and more.
- OkCupid has published their own analyses of this kind of data.

Our question: can we predict the sex of a user from their height?

What is wrong with this regression?

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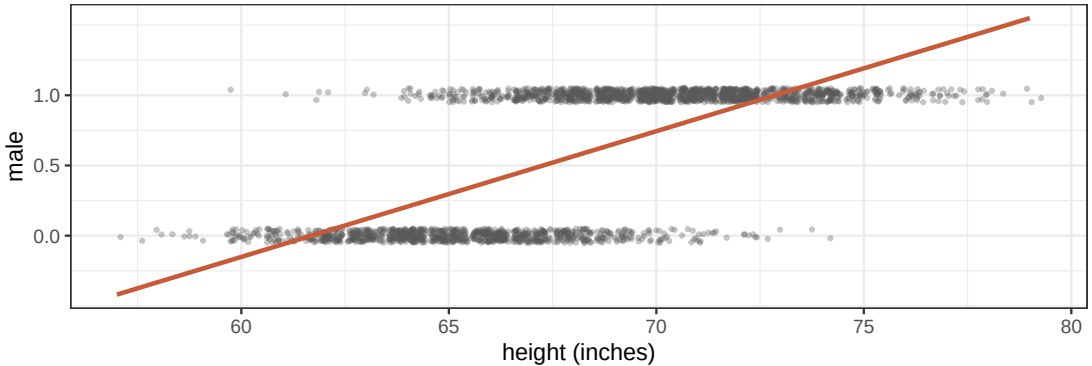
$$\widehat{\text{sex}} = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{height}$$

The Y variable is **categorical** — two levels, male or female.

Ordinary linear regression is built for a quantitative Y. Nothing stops us from running it, but we should ask what it would even mean.

But what if we just do it anyway?

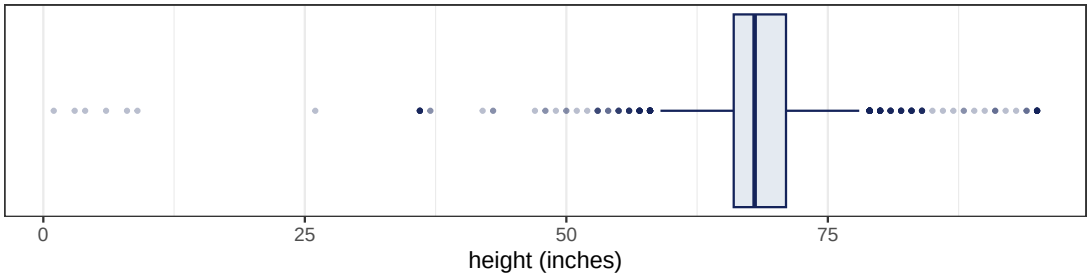
Code `male = 1`, `female = 0`, and fit a line:



A line is a spectacularly bad fit. And what would it mean to predict male $= 0.7$? Or 1.2?

First, clean the data

There are some impossible values for height:



Keep heights between 55 and 80 inches (4'7" and 6'8"). That drops only 120 of 59,946 cases.

```
my.profiles <- profiles %>% filter(height >= 55 & height <= 80)
```

The idea behind logistic regression

- Instead of predicting male/female, predict the **probability** that a user is male.
- One level is "success" ($Y = 1$), the other "failure" ($Y = 0$). Here male is success.
- Fit a curve that is always between 0 and 1.

To build that curve we need one piece of vocabulary first: **odds**.

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- In general, the odds of something with probability p are

$$\text{odds} = \frac{p}{1-p}$$

Odds: an example

During March Madness, a \$100 bet on San Diego State paid \$360 in winnings — odds to win of $1/3.6$.

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$$\frac{p}{1-p} = \frac{1}{3.6} \quad \Longrightarrow \quad p = \frac{1}{4.6} \approx 0.22$$

The logistic regression model

Logistic regression models the **log odds** of success as a linear function of X:

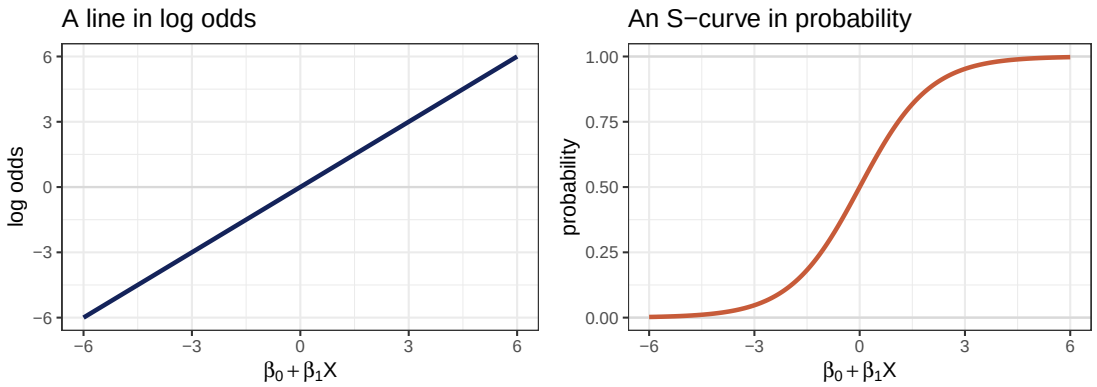
$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X$$

The left side runs over all of $(-\infty, \infty)$, so a straight line is a sensible thing to put on the right.

Solving back for p gives an S-shaped curve, bounded between 0 and 1:

$$p = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}}$$

The same model, two ways to look at it



Straight on the log-odds scale; squashed into $(0, 1)$ on the probability scale.

Let's fit it

```
model <- glm(male ~ height, data=my.profiles, family=binomial)
summary(model)


Call:
glm(formula = male ~ height, family = binomial, data = my.profiles)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -44.44861    0.35751    -124   <2e-16 ***
height        0.66190    0.00529     125   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

How do we read the fit?

The regression says

$$\log(\text{odds}) = \log\left(\frac{P(\text{male})}{1 - P(\text{male})}\right) = -44.45 + 0.66 \cdot \text{height}.$$

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Solving for the probability:

$$\widehat{P}(\text{male}) = \frac{e^{-44.45+0.66 \cdot \text{height}}}{1+e^{-44.45+0.66 \cdot \text{height}}}$$

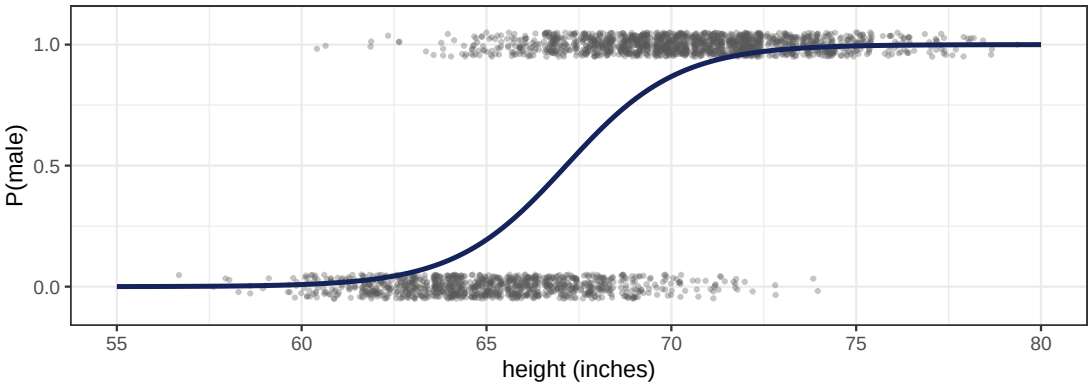
Making predictions

```
predict(model, data.frame(height=69), type="response")
```

```
     1
0.7725
```

Someone who is 5'9" has about a 77% chance of being male.

Visualizing the fitted model



The curve never leaves $[0, 1]$, and it bends where the data actually change over.

Interpreting the coefficients

$$\log\left(\frac{P(\text{male})}{1-P(\text{male})}\right) = -44.45 + 0.66 \cdot \text{height}$$

- At height $= 0$, the predicted log odds are -44.45, so the predicted probability is essentially 0.

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- Each extra inch of height adds 0.66 to the log odds.

A height of zero is nowhere near the data. The intercept anchors the line; it is not a claim about anybody.

Log odds are awkward. Odds are not.

Exponentiating both sides:

$$\widehat{\text{odds of male}} = e^{-44.45+0.66 \cdot \text{height}}$$

- One more inch **multiplies** the odds by $e^{0.6619} = 1.94$ — a 94% increase.

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Exponentiating both sides:

$$\widehat{\text{odds of male}} = e^{-44.45+0.66 \cdot \text{height}}$$

- One more inch **multiplies** the odds by $e^{0.6619} = 1.94$ — a 94% increase.
- Two more inches multiply the odds by $1.94^2 = 3.76$ — a 276% increase.

In linear regression slopes add. In logistic regression they multiply.

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- There are "pseudo-R^2" measures, but they are harder to interpret and easier to misuse.
- We will focus on something simple and concrete: **how many cases did we get right?**

Turning probabilities into predictions

Rule: predict male if $\widehat{P}(\text{male}) \geq 0.5$, otherwise female.

```
predicted.sex <- ifelse(predict(model, type="response") >= 0.5, "m", "f")
correct <- ifelse(predicted.sex == my.profiles$sex, 1, 0)
sum(correct) / nrow(my.profiles)

[1] 0.8309
```

83% of what?

Any accuracy number needs a baseline. What if we always guessed the majority?

```
xtabs(~ sex, data=my.profiles) %>% prop.table()

sex
     f      m
0.4027 0.5973
```

Always guessing male gets 60% right. Our model gets 83%.

That improvement is the **lift** the model buys us: from 60% to 83%.

The confusion matrix

Pick one category as **positive** (here, male) and the other as **negative** (female). Every case lands in one of four cells.

	Actual	
Predicted	Positive	Negative
Positive	TP	FP
Negative	FN	TN

FP = a woman predicted male. FN = a man predicted female.

The confusion matrix, for real

```
tab_conf <- xtabs(~ predicted.sex + my.profiles$sex)
tab_conf

             my.profiles$sex
predicted.sex     f     m
            f 19466  5494
            m  4623 30243
```

5,494 false negatives (men predicted female) and 4,623 false positives (women predicted male).

Careful reading the quadrants: the schematic above lists positive first, but R prints `f` first.

Rates from the confusion matrix

- **True positive rate** $= \mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$ — of the actual positives, the share we called positive
- **True negative rate** $= \mathrm{TN}/(\mathrm{TN}+\mathrm{FP})$ — of the actual negatives, the share we called negative
- **False positive rate** $= \mathrm{FP}/(\mathrm{TN}+\mathrm{FP})$
- **False negative rate** $= \mathrm{FN}/(\mathrm{TP}+\mathrm{FN})$

Each rate is a column of the confusion matrix, normalized.

The same table, normalized

```
xtabs(~ predicted.sex + my.profiles$sex) %>% prop.table(2)

             my.profiles$sex
predicted.sex      f      m
            f 0.8081 0.1537
            m 0.1919 0.8463
```

`prop.table(2)` divides by the column totals, so each column is one actual group: we catch 81% of the women and 85% of the men.

Evaluating a classifier: three cautions

- The 0.5 threshold is a **choice**. Move it and you trade false positives against false negatives.
- Those two errors rarely cost the same. A missed loan default and a wrongly denied loan are not equivalent; neither are a missed tumor and a false alarm.
- Everything here is in-sample, and therefore optimistic. For an honest number, hold out a test set or cross-validate.

Adding a predictor

Just like linear regression: each coefficient is the effect on the log odds, holding the others constant.

```
xtabs(~ orientation, data=my.profiles)

orientation
bisexual       gay straight
    2763      5568    51495
```

`orientation` has three categories, so R will fit two dummies against a reference level.

The fit

```
model2 <- glm(male ~ height + orientation, data=my.profiles,
              family=binomial)
summary(model2)


Call:
glm(formula = male ~ height + orientation, family = binomial,
    data = my.profiles)

Coefficients:
                   Estimate Std. Error z value Pr(>|z|)
(Intercept)         -46.08076    0.37167  -124.0   <2e-16 ***
height                0.66535    0.00537   124.0   <2e-16 ***
orientationgay        2.09556    0.07209    29.1   <2e-16 ***
orientationstraight   1.39972    0.06068    23.1   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpreting the coefficients

$$\log\left(\frac{p}{1-p}\right) = -46.08 + 0.67 \cdot \text{height} + 2.10 \cdot \text{gay} + 1.40 \cdot \text{straight}$$

- The intercept is the log odds for a **bisexual** user of height 0 — the reference level.

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- Within a given orientation, each extra inch multiplies the odds by $e^{0.6653} = 1.95$.

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- The intercept is the log odds for a **bisexual** user of height 0 — the reference level.
- Within a given orientation, each extra inch multiplies the odds by $e^{0.6653} = 1.95$.
- At the same height, gay vs. bisexual multiplies the odds by $e^{2.0956} = 8.13$; straight vs. bisexual, by $e^{1.3997} = 4.05$.

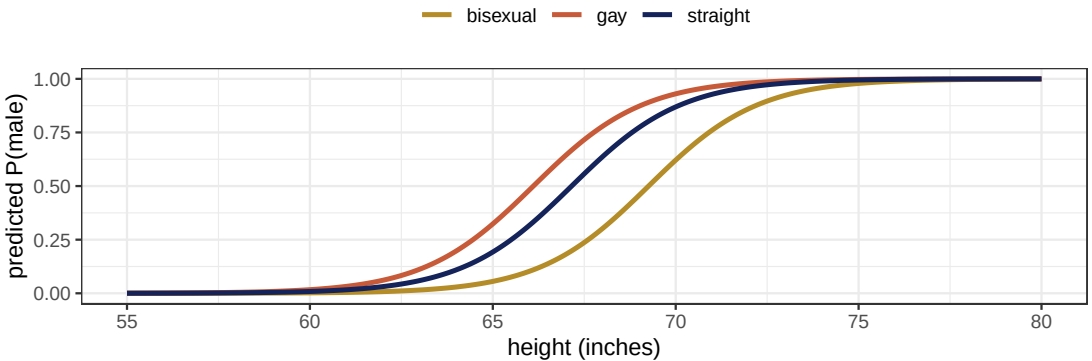
Why those dummies are so large

```
xtabs(~ sex + orientation, data=my.profiles) %>% prop.table(2)

   orientation
sex bisexual    gay straight
  f   0.7217 0.2848   0.3983
  m   0.2783 0.7152   0.6017
```

The bisexual profiles are 72% female; the gay profiles are 72% male. Orientation carries real information about sex in this data, quite apart from height.

Three curves, same slope



Adding a categorical predictor **shifts** the curve sideways. It does not change its steepness.

In linear regression, an interaction means the **slope** of X_1 depends on the value of X_2.

Exactly the same idea here: the effect of height on the odds is allowed to differ by orientation.

$$\texttt{male} \sim \texttt{height} * \texttt{orientation}$$

The interaction fit

```
int.model <- glm(male ~ height * orientation, data=my.profiles,
                 family=binomial)
summary(int.model)


Call:
glm(formula = male ~ height * orientation, family = binomial,
    data = my.profiles)

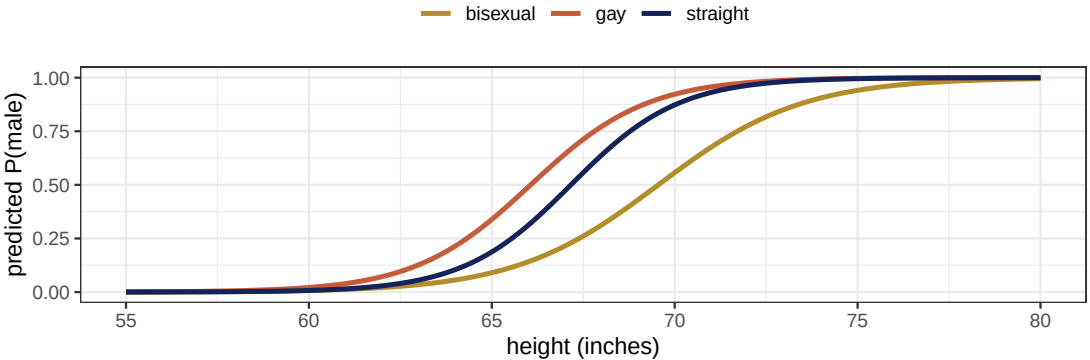
Coefficients:
                          Estimate Std. Error z value Pr(>|z|)
(Intercept)               -35.3027     1.4050  -25.13  < 2e-16 ***
height                      0.5076     0.0206   24.67  < 2e-16 ***
orientationgay             -6.2727     1.8365   -3.42  0.00064 ***
orientationstraight       -10.2887     1.4596   -7.05  1.8e-12 ***
height:orientationgay       0.1218     0.0271    4.49  7.1e-06 ***
height:orientationstraight  0.1712     0.0214    8.01  1.2e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The interaction model

$$\log\left(\frac{p}{1-p}\right) = -35.30 + 0.51 \cdot \text{height} \\ - 6.27 \cdot \text{gay} - 10.29 \cdot \text{straight} \\ + 0.12 \cdot \text{height} \cdot \text{gay} + 0.17 \cdot \text{height} \cdot \text{straight}$$

Each orientation now gets its own slope: 0.51 for bisexual, $0.51 + 0.12 = 0.63$ for gay, $0.51 + 0.17 = 0.68$ for straight.

Three curves, three slopes



Now the curves differ in steepness too: height is a sharper signal for straight users than for bisexual ones.

Three hypotheses worth testing

1. **Overall:** $\beta_1 = \beta_2 = \cdots = 0$. The model has no predictive power at all.

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1. **Overall:** $\beta_1 = \beta_2 = \cdots = 0$. The model has no predictive power at all.
2. **One quantitative predictor:** $\beta_i = 0$. That X has no effect, holding the others constant.
3. **One dummy:** $\beta_i = 0$. That category is no different from the reference level, holding the others constant.

Testing the overall null

The **likelihood ratio test** compares the full model against an intercept-only model.

For `male ~ height + orientation`, the test statistic is $\chi^2 = 36{,}932$ on 3 degrees of freedom, with $p < 0.001$.

With 59,826 profiles, this is not a close call — but note that with a sample this large, almost any non-zero effect will be "significant."

For one quantitative predictor (height) or one dummy (straight vs. bisexual), read the z value and p-value straight off the `summary` table.

	Estimate	Std. Error	z	$\Pr(>\|z\|)$
`height`	0.6653	0.0054	124.0	$< 2 \times 10^{-16}$
`orientationgay`	2.0956	0.0721	29.1	$< 2 \times 10^{-16}$
`orientationstraight`	1.3997	0.0607	23.1	$< 2 \times 10^{-16}$

Same logic as the t tests in linear regression; the reference distribution is normal, not t.

What else is logistic regression for?

- **Finance:** which customers are likely to default on a loan
- **Medicine:** whether a patient has a condition, given test results
- **Advertising:** who will respond to a campaign
- **Marketing:** who will purchase, subscribe, or sign up

Any time the answer is yes or no and you want a probability rather than a guess.

What to take away

- Model the **log odds** as linear; read the fit back as an S-curve in probability.
- Coefficients **multiply the odds**: e^{β} is the interpretable number.
- Evaluate with a **confusion matrix** against a baseline, not with R^2.
- The 0.5 threshold is yours to move, and the two kinds of error rarely cost the same.